

# D1: identical current pixels hide incompatible viable actions

Previous frames reveal motion; a memoryless current frame merges the roots.

**Root 1: incoming left**

hidden  $v = -1.85$  | viable  $a = \{+1\}$

**Root 2: incoming right**

hidden  $v = +3.85$  | viable  $a = \{-1\}$

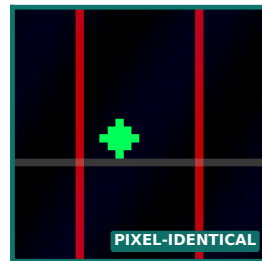
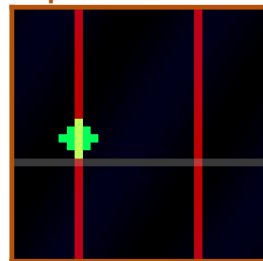
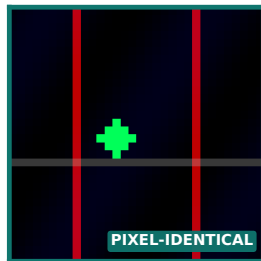
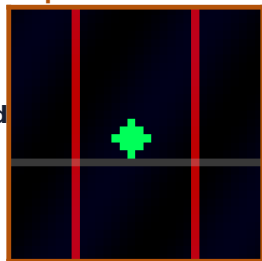
previous frame

current frame

previous frame

current frame

Controlled  
cart



Current pixels: one shared code | Previous pixels: two distinct histories | Common viable first actions: empty

**Root 1: clockwise**

hidden  $\omega = -3.00$  | viable  $a = \{+1\}$

**Root 2: counterclockwise**

hidden  $\omega = +3.25$  | viable  $a = \{-1, 0\}$

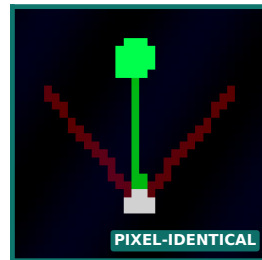
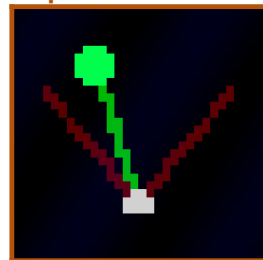
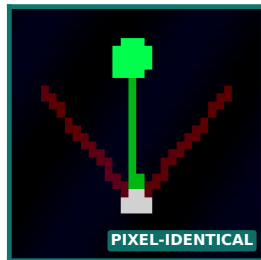
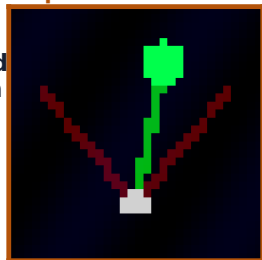
previous frame

current frame

previous frame

current frame

Controlled  
pendulum



Current pixels: one shared code | Previous pixels: two distinct histories | Common viable first actions: empty

Registered three-step, noise-free finite-action trees; hidden velocities are shown for interpretation only.